QP CODE: 25900015

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP(HONOURS) REGULAR EXAMINATION, MARCH 2025
SECOND SEMESTER
DISCIPLINE SPECIFIC CORE COURSE(DSC)-MG2DSCBTS100-PREPARATION OF BIOLOGICAL SPECIMENS
(*2024 Admission onwards)*

DURATION: 1.5 MARKS MAX.MARKS: 50

PART A

Multiple Choice Questions
Answer **All** questions. Each questions carries **1** mark

1. **What is a transect in ecological samplings? U /(CO1)**
   Ans: A straight line or path along which observations are made.
2. **Why is glycerin sometimes used in fish specimen preservation? U/ (CO1, CO2)**
   Ans: To prevent drying
3. **Which of the following is an example of a simple stain? K /(CO4)**
   Ans: Methylene Blue
4. **Why is it called an "orange peel" bucket? K /(CO1)**
   Ans: its design resembles the opening and closing of orange peels
5. **Which type of mounting media is commonly used for preparing permanent slides**?
   Ans: Canada balsam **U/(CO4)**
6. **Which of the following is a reason for using an articulated skeleton in museums or classrooms? K /(CO3)**
   Ans: All of the Above
7. **The respiratory organs in arthropods can be: U/ (CO1, CO2)**
   Ans: All of the Above
8. **Why is tissue embedding essential before using a rotary microtome. U/(CO3)**
   Ans: To provide structural support and stability for sectioning
9. **Which of the following organisms belongs to the phylum coelenterate? U/ (CO1, CO2)**
   Ans: Hydra
10. **Which complex tissue is responsible for the transport of water and minerals in plants?**
    Ans: Xylem **U / (CO4)**
11. **Which of the following is an example of a parasitic nematode? U / (CO1, CO2)**
    Ans: Ascaris
12. **What is the primary excretory organ in Mollusca? U/ (CO1, CO2)**
    Ans: Nephridia

(12 x 1 = 12 Marks)

Part B

Short answer questions
Answer any **6** questions. Each question carries **3** marks

13. **What are the main stages in the life cycle of frog? K /(CO3)**
    Ans: The process typically takes around 14 weeks.
    1**. Egg (Frogspawn):** Female frogs lay their eggs, often in water, in large clusters known as frogspawn,contains a yolk that provides nourishment for the developing tadpole. **2. Tadpole:** Hatch from the eggs, initially resembling fish with gills and tails, live in water and feed on plant matter and vegetation. **3. Froglet**: As tadpoles develop, they start to grow legs, their tails shrink, and their gills begin to transform into lungs.**4. Adult Frog:** Froglets eventually become adult frogs, capable of living on land and returning to water to reproduce. Adult frogs have lost their tails and can breathe through their lungs.

14. **Explain the importance of fixation in Museum specimen preparation? U /(CO5)**
Ans: Fixation in museum specimen preparation is crucial for preserving tissue structure, preventing decay, and enabling long-term display and study. It stops autolysis, kills microorganisms, and hardens the specimen while maintaining a "life-like" state, allowing for subsequent processing and analysis. Fixatives like formalin are used to achieve these goals.

15. **Name two methods of specimen preservation. K/(CO1)**
Ans: Two common methods of specimen preservation are liquid preservation (using chemicals like formalin or alcohol) and dry preservation (like herbarium specimens, taxidermy). Liquid preservation is used to maintain the integrity of soft tissues, while dry preservation focuses on drying plant specimens.

16. **Who is responsible for enforcing specimen collection laws? K /(CO1)**
Ans: The enforcement of specimen collection laws typically falls under various governmental and regulatory authorities, depending on the location and type of specimen.
Main objectives:
- Protect endangered or threatened species
- Ensure sustainable collection practices
- Prevent habitat destruction
- Maintain biodiversity records
- Regulate international transport and trade
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17. **What are the two main groups of angiosperms based on their seed structure and give an example of each? K/(CO4)**
Ans: Angiosperms are broadly classified into two groups based on seed structure: monocots and dicots. Monocots have one seed leaf (cotyledon) in their seed, while dicots have two. Examples of monocots include corn, wheat, and lilies, while examples of dicots include roses, sunflowers, and tomatoes.

18. **What is the role of airtight containers in wet preserved plant storage? U / (CO5)**
Ans: Airtight containers play a crucial role in wet preserved plant storage by:

1. **Preventing Evaporation**: They stop the preservative solution (e.g., formalin, alcohol) from evaporating, which helps maintain the correct concentration needed to preserve the plant tissue.
2. **Avoiding Contamination**: Airtight sealing prevents the entry of microorganisms, dust, or air that could lead to decomposition or fungal/bacterial growth.
3. **Maintaining Chemical Stability**: They limit exposure to air and light, which can degrade some preservatives over time.
4. **Ensuring Safety**: Prevents the release of harmful fumes from chemicals like formaldehyde, protecting users from toxic exposure.

19. **What adaptation helps reptiles survive in terrestrial environments? U / (CO1, CO2)**
Ans: Reptiles have several key adaptations that allow them to thrive in terrestrial environments.
Scaly, waterproof skin – This adaptation reduces water loss through evaporation, allowing reptiles to live in dry, land-based habitats without drying out.

- **Amniotic eggs with leathery shells**, which can develop on land without needing water.
- **Efficient lungs** for breathing air.
- **Ectothermic metabolism**, allowing them to regulate body temperature through behavior (like basking in the sun).

20. **How does the season affect plant collection? U / (CO5)**
Ans: The season significantly affects plant collection due to changes in plant growth, reproduction, and availability. Here's how:

1. **Flowering and Fruiting**: Many plants can only be properly identified when they have flowers or fruits, which appear in specific seasons (usually spring or summer).
2. **Leaf Condition**: In dry or cold seasons, leaves may wilt, fall, or change color, making collection and identification harder.
3. **Plant Availability**: Some annual or seasonal plants may not be visible or accessible during off-seasons (like winter or dry periods).
4. **Preservation Quality**: Plants collected in optimal seasons are usually healthier and easier to preserve.

(6 x 3 =18 Marks)

Part C

Short Essay Questions

Answer any **4** questions. Each question carries **5** mark**s.**

21. What are biological specimen preservatives and why are they necessary for research and study? U / (CO1)

Ans: Biological specimen preservatives are chemicals or solutions used to prevent the decay and decomposition of biological tissues (like plant, animal, or microbial samples) after collection. Common examples include formalin, alcohol (ethanol), glycerin, and FAA (formalin-acetic acid-alcohol).

Reason: Prevent Decomposition. Maintain Structural Integrity. Allow Long-term Storage. Enable Detailed Study.

22. Explain how taxidermy is used in the preparation of mammal specimens. U / (CO1, CO2)

Ans: Taxidermy is a method used to preserve and display the external appearance of animals, especially mammals, for scientific study, education, or exhibition.

Steps in Taxidermy for Mammals:

1. Skinning the Animal
2. Preserving the Skin
3. Creating a Mount
4. Mounting the Skin.
5. Detailing

23. **What precautions should be taken while storing herbarium sheets to prevent damage? K/ (CO5)**

Ans: To prevent damage, the following **precautions should be taken while storing herbarium sheets**:

- Keep in a Dry Environment: Moisture can cause fungal growth and decay, so store sheets in a cool, dry place with low humidity.
- Use Insect Repellents: Place naphthalene balls or pest-repelling chemicals to protect from insects like silverfish and beetles.
- Avoid Direct Sunlight: Store away from light to prevent fading of plant parts and labels.
- Use Acid-Free Paper: Mount specimens on acid-free herbarium sheets to prevent chemical damage over time.
- Label Clearly and Securely: Ensure labels are complete, typed or clearly handwritten, and firmly attached.
- Flat, Covered Storage: Store sheets flat in closed cabinets or boxes to avoid bending, dust, and physical damage.
- Regular Inspection: Check periodically for signs of pests, mold, or physical wear and take prompt action if needed.
- Handle with Care: Use clean, dry hands or gloves and support sheets from beneath to avoid tears or breakage.

24. What challengers do researchers face while collecting plant specimens in remote area? **U/ (CO5)**

Ans: Researchers face several **challenges while collecting plant specimens in remote areas**, including:

1. **Difficult Terrain**: Mountains, dense forests, or swamps can be physically hard to access and navigate.
2. **Harsh Weather Conditions**: Extreme heat, rain, or cold can hinder movement and damage equipment or specimens.
3. **Limited Supplies**: Lack of access to preservation materials, tools, or emergency supplies can affect specimen quality and safety.
4. **Health Risks**: Exposure to wild animals, insects, poisonous plants, or diseases (like malaria) poses danger.
5. **Communication Barriers**: Poor or no network connectivity makes coordination and emergency communication difficult.
6. **Legal and Permitting Issues**: Some areas require special permissions or have restrictions on collecting specimens.
7. **Transport Difficulties**: Bringing collected specimens back safely without damage can be challenging due to lack of proper packaging or transport options.
8. **Data Recording Problems**: Harsh conditions can make it hard to write or document details properly in the field. (Any **5** points)

25. Briefly classify phylum porifera and provide suitable examples for each group. **K / (CO1, CO2)**

Ans: Phylum Porifera (sponges) is classified based on the composition of their skeletons and canal systems. They are simple, aquatic, multicellular animals with pores and no true tissues. Porifera is mainly divided into three classes: Class Calcarea (Calcareous Sponges), Class Hexactinellida (Glass Sponges) and Class Demospongiae (Bath Sponges).

Examples:

Class Clacarea- Sycon, Leucosolenia

Class Hexactinellida- Venus' flower basket

Class Demospongiae- Spongilla (freshwater sponge), Euspongia (bath sponge)

(Introduction 1marks; classification-2 marks; examples-2 marks)

26. What is the process of embedding biological specimens in resin and what material are commonly used? **K / (CO3)**

Ans: Embedding biological specimens in resin is a preservation technique used to protect delicate samples (like insects, plant parts, or small animals) in a clear, solid block for long-term storage, study, or display. This process keeps the specimen visible and prevents physical damage, decay, or contamination.

Process of Embedding in Resin:

1. **Fixation**: The specimen is preserved using chemicals like formalin or alcohol to stop decay.
2. **Dehydration**: Water is removed from the specimen using increasing concentrations of alcohol.
3. **Positioning**: The specimen is arranged in a mold (plastic or silicone) in the desired pose.
4. **Resin Pouring**: Clear liquid resin is poured over the specimen in layers to avoid bubbles.
5. **Curing**: The resin hardens (cures) either at room temperature or with heat/UV light, depending on the resin type.
6. **Finishing**: Once fully hardened, the resin block is removed from the mold and polished for clarity.

Commonly Used Embedding Materials: Epoxy Resin, Polyester Resin, Acrylic Resin (PMMA) and Silicone Molds. (Introduction -1 marks; steps-3 marks; materials-1 marks).

(4 x 5 = 20 Marks)